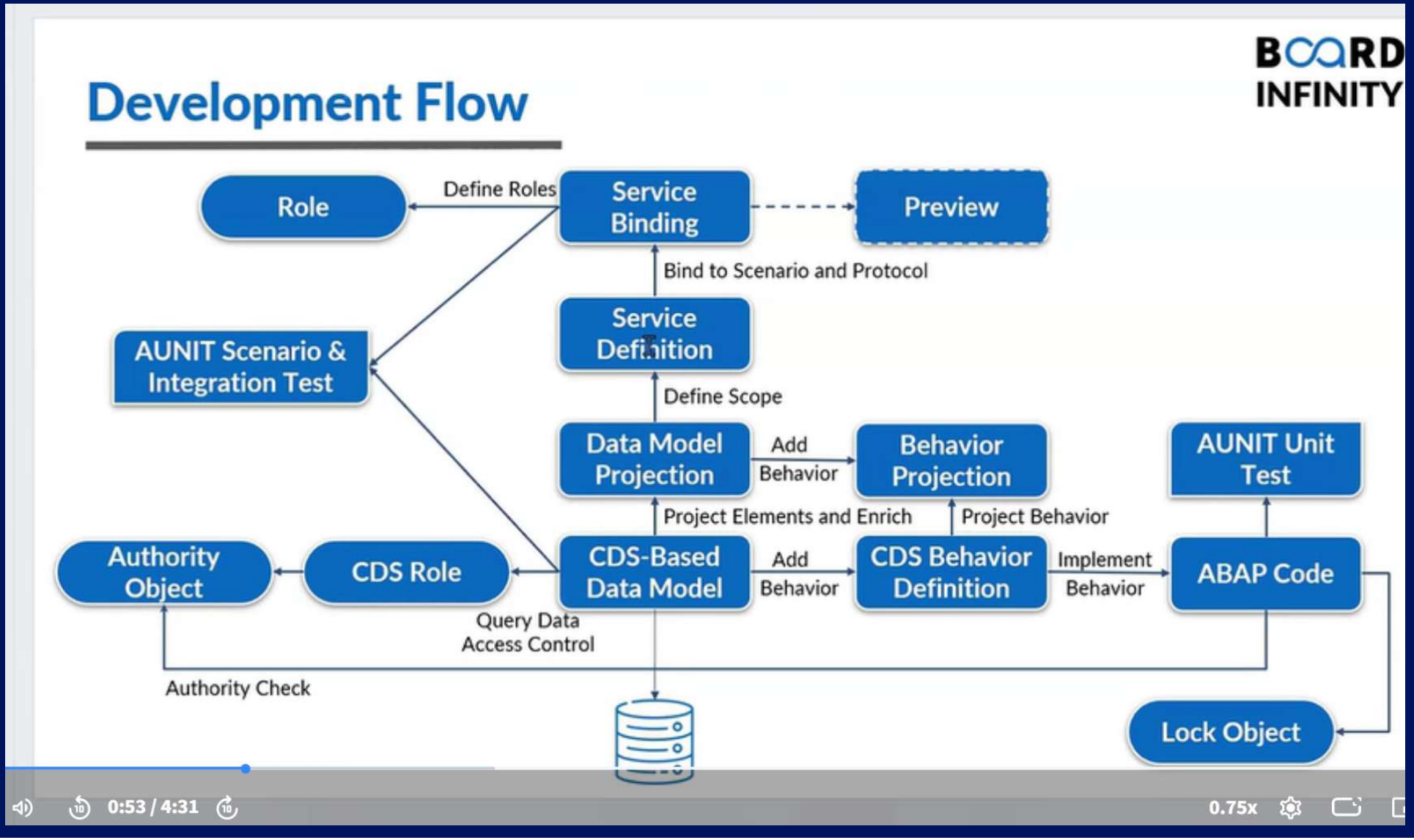


ABAP RESTFUL PROGRAMMING MODEL (RAP) AND EXTENSIONS

COURSERA ONLINE COURSE

MANAGED SCENARIO



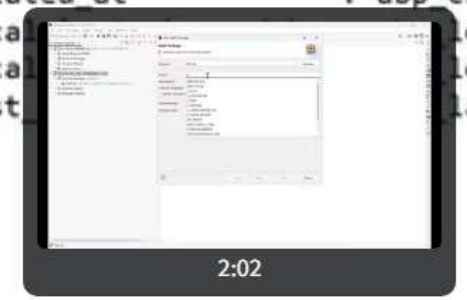
Create Package and Database Table

- Create a package to organize RAP objects.
- Define a database table for persistent data.
- Include fields like TravelID, Agency ID, CustomerID, BeginDate, EndDate, Total Price.
- Ensure table has primary key and necessary fields for Travel app.
- Save (Ctrl+S) and activate (Ctrl+F3).

```
@EndUserText.label : 'Travel Data'
@AbapCatalog.enhancement.category : #NOT_EXTENSIBLE
@AbapCatalog.tableCategory : #TRANSPARENT
@AbapCatalog.deliveryClass : #A
@AbapCatalog.dataMaintenance : #RESTRICTED
define table zbi_rap_atrav {

  key client                : abap.clnt not null;
  key travel_id             : /dmo/travel_id not null;
  agency_id                 : /dmo/agency_id;
  customer_id               : /dmo/customer_id;
  begin_date                : /dmo/begin_date;
  end_date                  : /dmo/end_date;
  @Semantics.amount.currencyCode : 'zbi_rap_trav.currency_code'
  booking_fee               : /dmo/booking_fee;
  @Semantics.amount.currencyCode : 'zbi_rap_trav.currency_code'
  total_price               : /dmo/total_price;
  currency_code             : /dmo/currency_code;
  description               : /dmo/description;
  overall_status            : /dmo/overall_status;
  attachment                : /dmo/attachment;
  mime_type                 : /dmo/mime_type;
  file_name                 : /dmo/filename;
  created_by                : abp_creation_user;
  created_at                : abp_creation_tstmpl;
  loca                      locinst_lastchange_user;
  loca                      locinst_lastchange_tstmpl;
  last                      lastchange_tstmpl;


}
```



Create Data Generator Class

- Create an ABAP class to populate table.
- Implement logic to insert sample data into Persistent table
- Use INSERT statements or internal tables for data generation.
- Execute class to fill database table with test data.
- Example: Insert 10 travel records with random dates and prices.

```
CLASS zst29bi_cl_rap_gen_data DEFINITION
PUBLIC
FINAL
CREATE PUBLIC .
PUBLIC SECTION.
INTERFACES if_oo_adt_classrun.
PROTECTED SECTION.
PRIVATE SECTION.
ENDCLASS.

CLASS zst29bi_cl_rap_gen_data IMPLEMENTATION.
DELETE FROM zst29_travel.
INSERT zst29_travel FROM (
SELECT
FROM /dmo/travel AS travel
FIELDS
travel~travel_id AS travel_id,
travel~agency_id AS agency_id,
travel~customer_id AS customer_id,
travel~begin_date AS begin_date,
travel~end_date AS end_date,
travel~booking_fee AS booking_fee,
travel~total_price AS total_price,
METHOD if_oo_adt_classrun~main.
DATA:
attachment TYPE /dmo/attachment,
file_name  TYPE /dmo/filename,
mime_type  TYPE /dmo/mime_type.
```

```
DELETE FROM zst29_travel.
INSERT zst29_travel FROM (
SELECT
FROM /dmo/travel AS travel
FIELDS
travel~travel_id AS travel_id,
travel~agency_id AS agency_id,
travel~customer_id AS customer_id,
travel~begin_date AS begin_date,
travel~end_date AS end_date,
travel~booking_fee AS booking_fee,
travel~total_price AS total_price,
```

```
travel~currency_code AS currency_code,
travel~description AS description,
CASE travel~status
WHEN 'N' THEN 'O'
WHEN 'P' THEN 'O'
WHEN 'B' THEN 'A'
ELSE 'X'
END AS overall_status,
@attachment AS attachment,
@mime_type AS mime_type,
@file_name AS file_name,
travel~createdby AS created_by,
travel~createdat AS created_at,
travel~lastchangedby AS last_changed_by,
travel~lastchangedat AS last_changed_at,
travel~lastchangedat AS local_last_changed_at
ORDER BY travel_id UP TO 10 ROWS
).
COMMIT WORK.
out->write( |[RAP100] Demo data generated for table | ).
ENDMETHOD.

ENDCLASS.
```

Run as abap application an created data may be seen in the table :



Data Preview

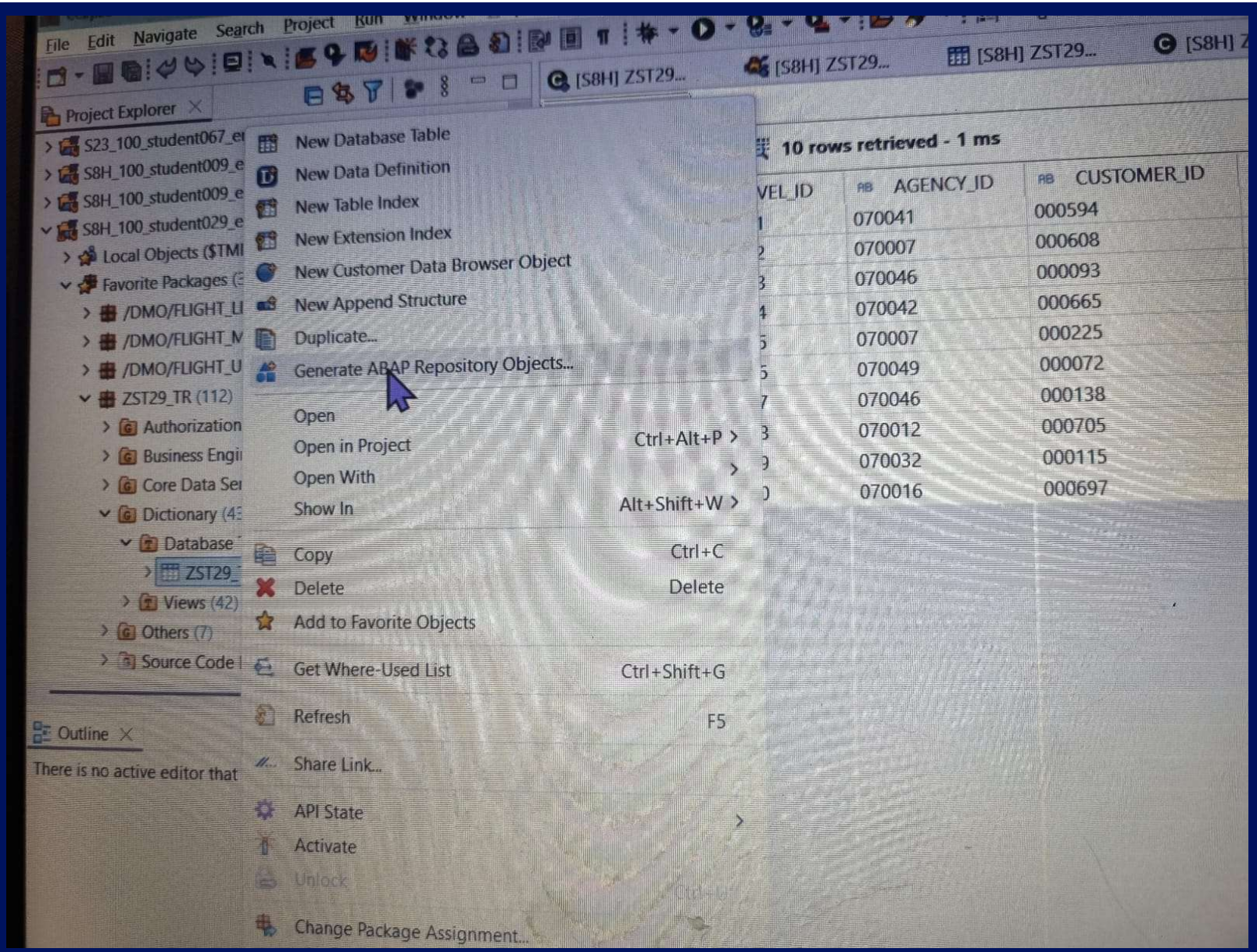
Show Log | Max. Rows: 100

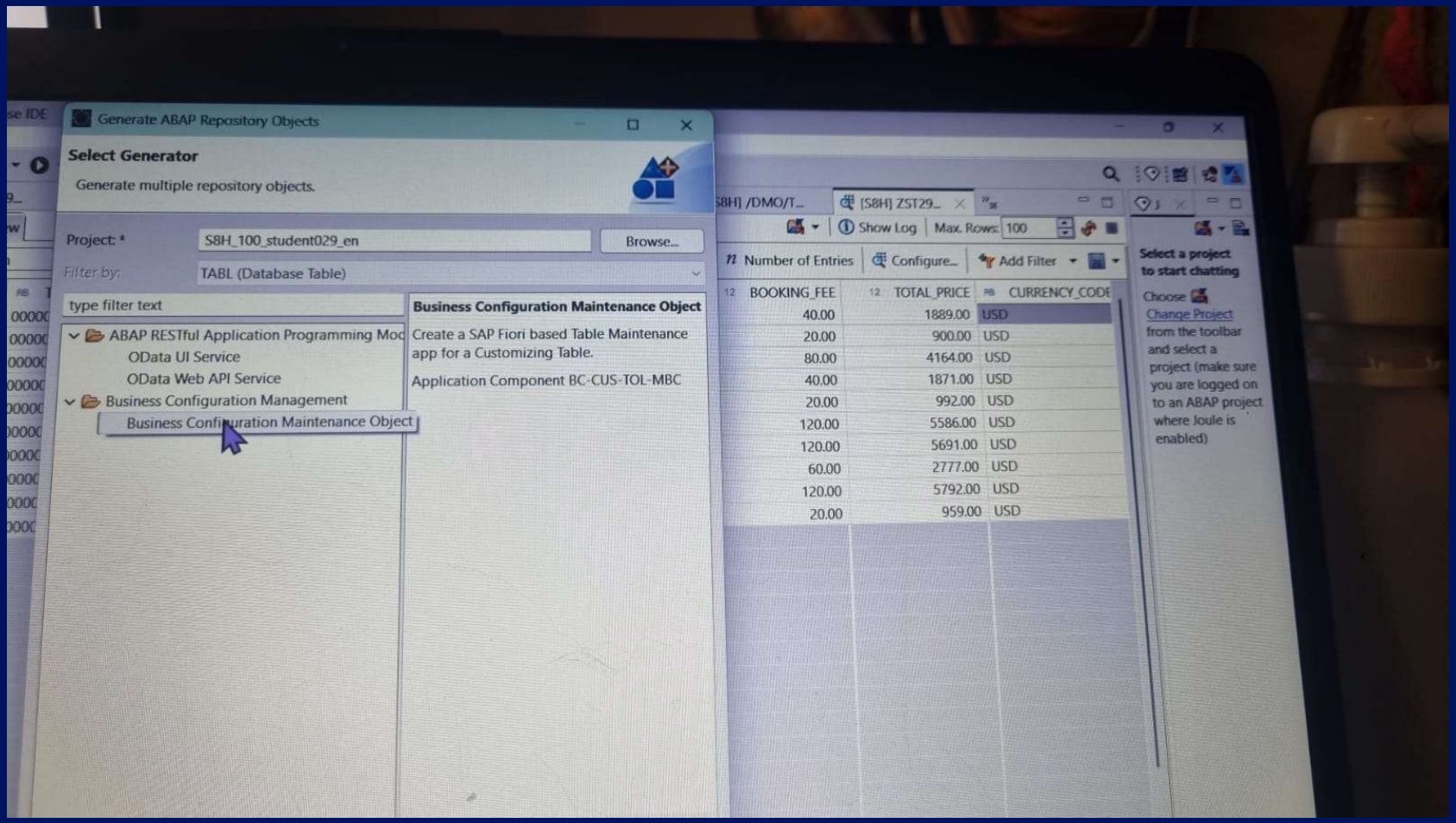
find pattern | 10 rows retrieved - 1 ms | SQL Console | Data Aging | Number of Entries | Configure... | Add Filter

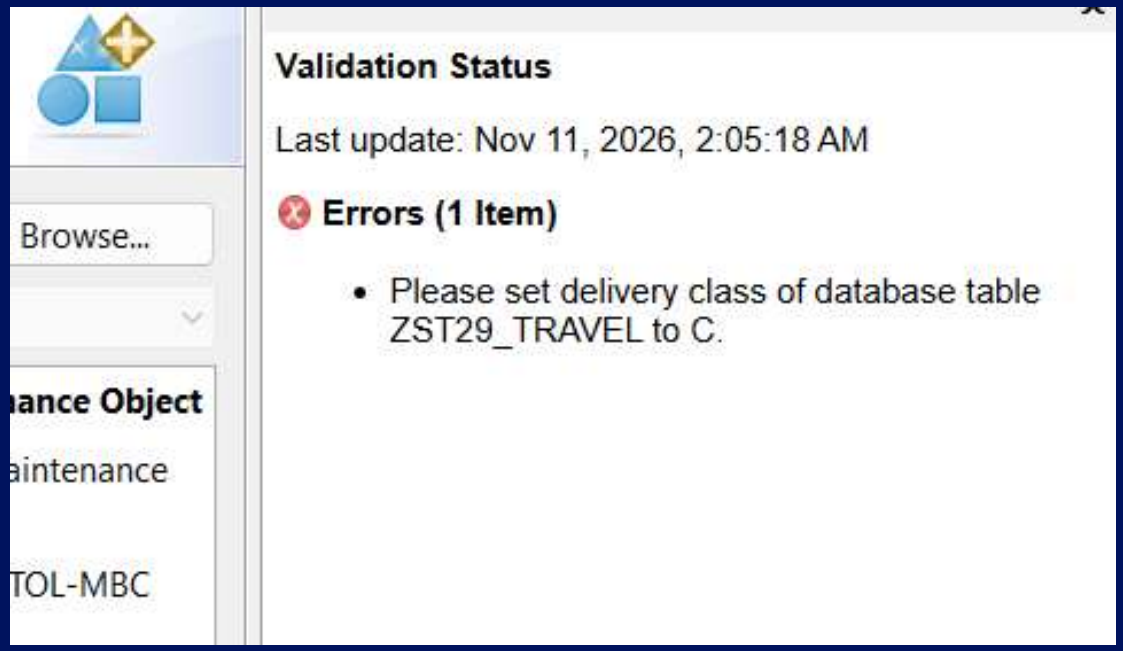
CLIENT	TRAVEL_ID	AGENCY_ID	CUSTOMER_ID	BEGIN_DATE	END_DATE	BOOKING_FEE	TOTAL_PRICE	CURRENCY_CODE	DESCRIPTION	OV
100	00000001	070041	000594	2021-04-03	2022-01-30	40.00	1889.00	USD	Vacation	O
100	00000002	070007	000608	2021-04-03	2021-04-03	20.00	900.00	USD	Business Trip for Ch...	O
100	00000003	070046	000093	2021-04-03	2022-01-30	80.00	4164.00	USD	Vacation	O
100	00000004	070042	000665	2021-04-03	2022-01-30	40.00	1871.00	USD	Vacation	O
100	00000005	070007	000225	2021-04-03	2021-04-03	20.00	992.00	USD	Business Trip for Ku...	O
100	00000006	070049	000072	2021-04-03	2022-01-30	120.00	5586.00	USD	Vacation	O
100	00000007	070046	000138	2021-04-03	2022-01-30	120.00	5691.00	USD	Vacation	O
100	00000008	070012	000705	2021-04-03	2021-04-05	60.00	2777.00	USD	Vacation for Lothar,...	O
100	00000009	070032	000115	2021-04-03	2022-01-30	120.00	5792.00	USD	Visiting Anna	A
100	00000010	070016	000697	2021-04-03	2021-04-05	20.00	959.00	USD	Business Trip to USA	O

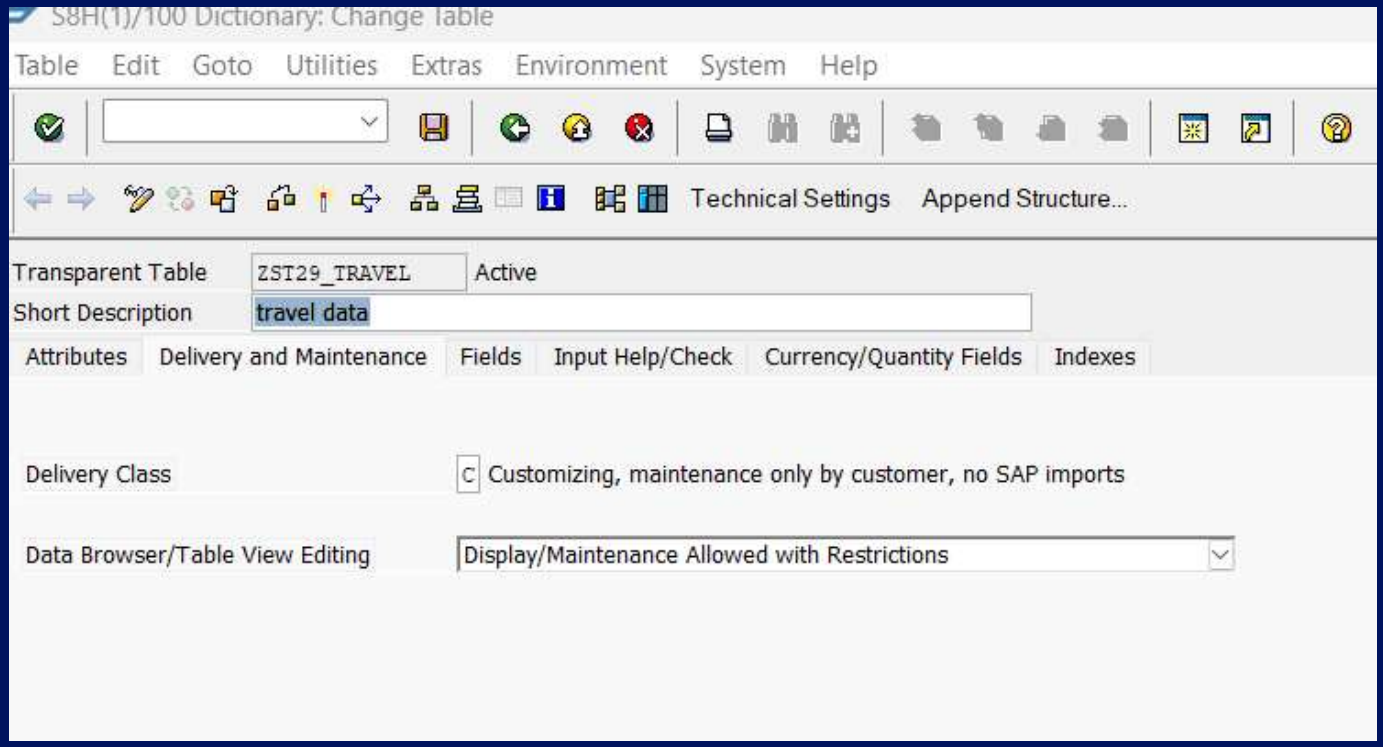
Create Business Configuration Management

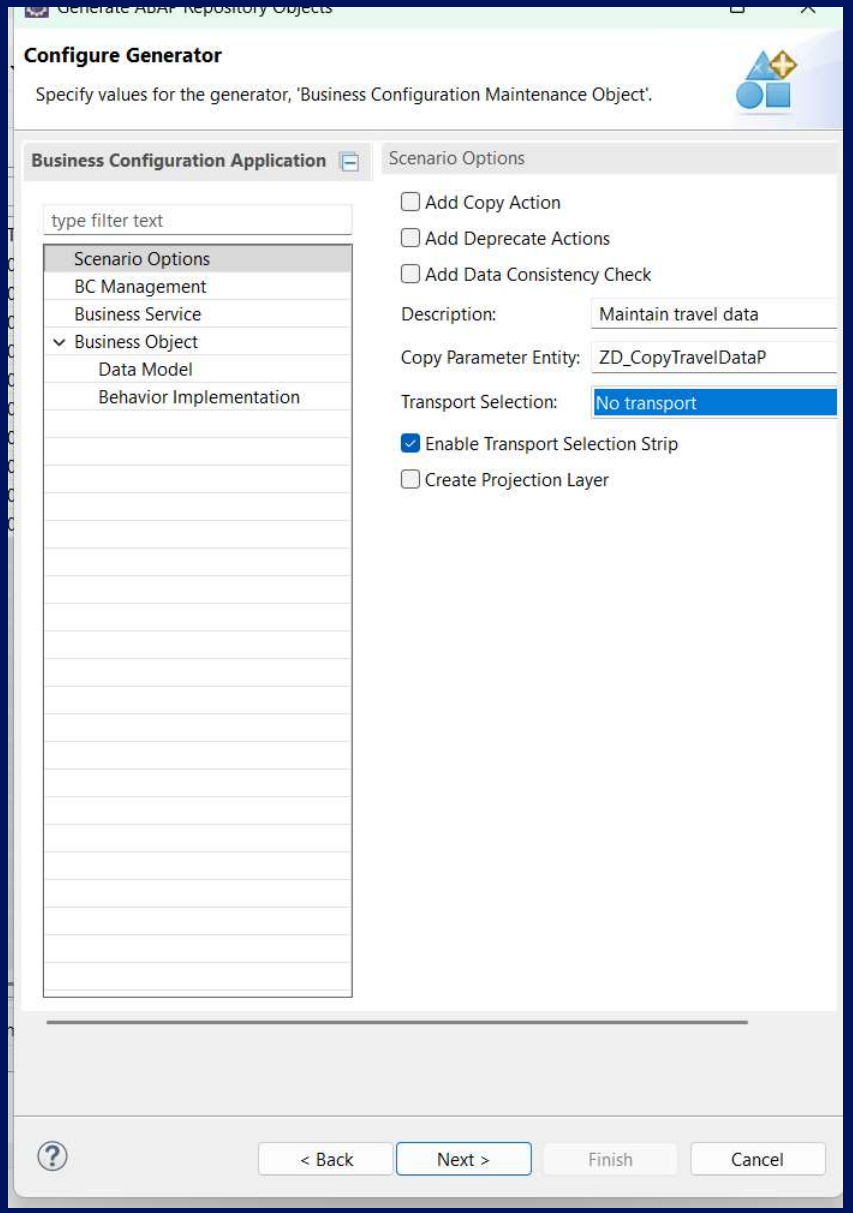
- Define business configuration in CDS view for app settings.
- Create a configuration table for parameters.
- Use CDS annotations to expose configuration in Fiori app.
- Enable maintenance via Business Configuration Maintenance app.
- Test configuration in Fiori Elements UI.

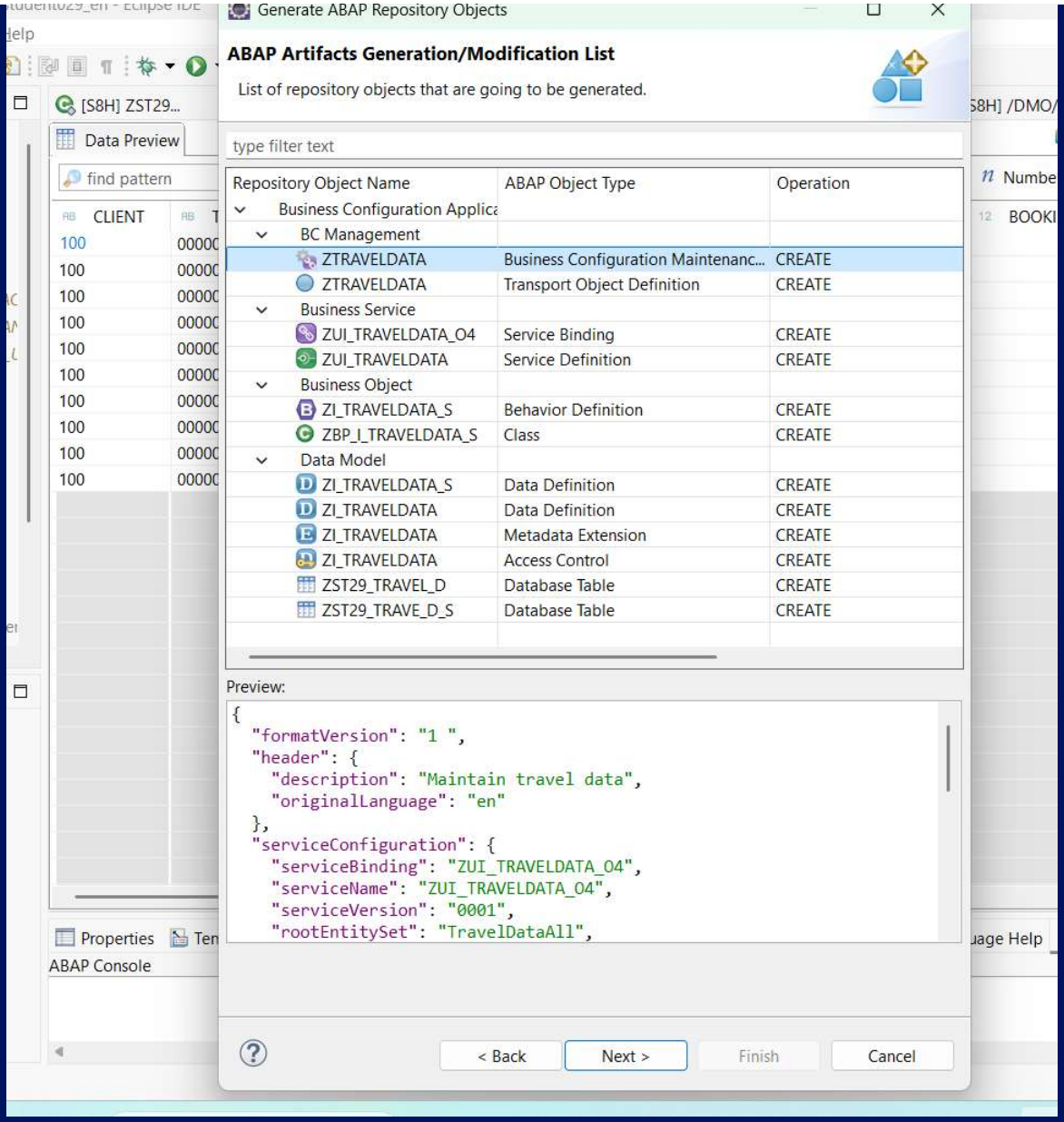

File Edit Navigate Search Project Run
Project Explorer
[S8H] ZST29...
[S8H] ZST29...
[S8H] ZST29...
S23_100_student067_e
S8H_100_student009_e
S8H_100_student009_e
S8H_100_student029_e
Local Objects ($TMI
Favorite Packages
/DMO/FLIGHT_LI
/DMO/FLIGHT_M
/DMO/FLIGHT_U
ZST29_TR (112)
Authorization
Business Engi
Core Data Se
Dictionary (4
Database
ZST29_
Views (42)
Others (7)
Source Code
Outline
There is no active editor that
New Database Table
New Data Definition
New Table Index
New Extension Index
New Customer Data Browser Object
New Append Structure
Duplicate...
Generate ABAP Repository Objects...
Open
Open in Project Ctrl+Alt+P
Open With
Show In Alt+Shift+W
Copy Ctrl+C
Delete Delete
Add to Favorite Objects
Get Where-Used List Ctrl+Shift+G
Refresh F5
Share Link...
API State
Activate
Unlock
Change Package Assignment...
10 rows retrieved - 1 ms
VEL_ID
AGENCY_ID
CUSTOMER_ID
070041 000594
070007 000608
070046 000093
070042 000665
070007 000225
070049 000072
070046 000138
070012 000705
070032 000115
070016 000697

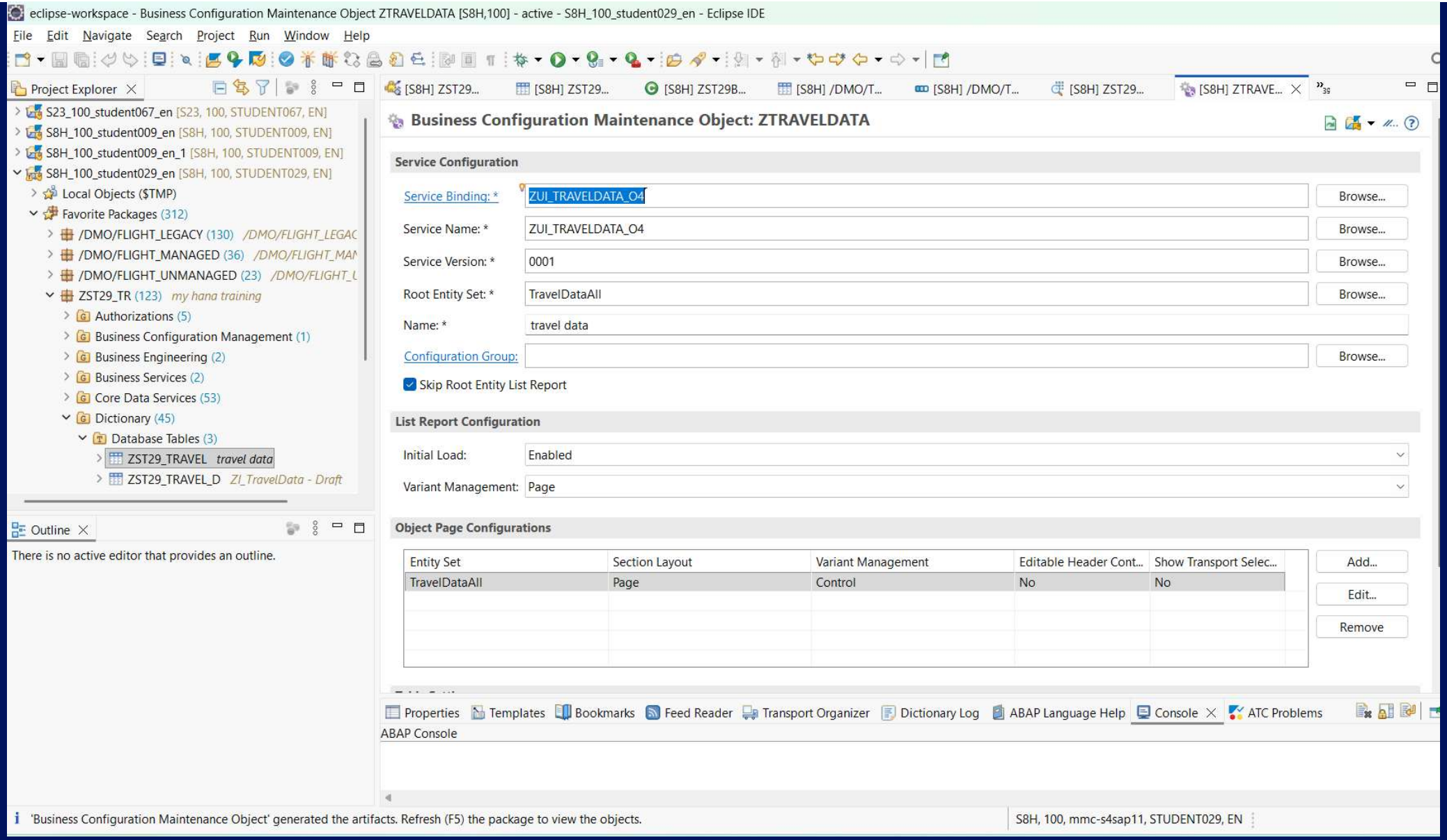

Generate ABAP Repository Objects
Select Generator
Generate multiple repository objects.
Project: *
S8H_100_student029_en
Browse...
Filter by:
TABL (Database Table)
type filter text
ABAP RESTful Application Programming Mod
OData UI Service
OData Web API Service
Business Configuration Management
Business Configuration Maintenance Object
Business Configuration Maintenance Object
Create a SAP Fiori based Table Maintenance app for a Customizing Table.
Application Component BC-CUS-TOL-MBC
Show Log
Max. Rows: 100
Number of Entries
Configure...
Add Filter
BOOKING_FEE
TOTAL_PRICE
CURRENCY_CODE
40.00 1889.00 USD
20.00 900.00 USD
80.00 4164.00 USD
40.00 1871.00 USD
20.00 992.00 USD
120.00 5586.00 USD
120.00 5691.00 USD
60.00 2777.00 USD
120.00 5792.00 USD
20.00 959.00 USD
Select a project to start chatting
Choose Change Project from the toolbar and select a project (make sure you are logged on to an ABAP project where Joule is enabled)

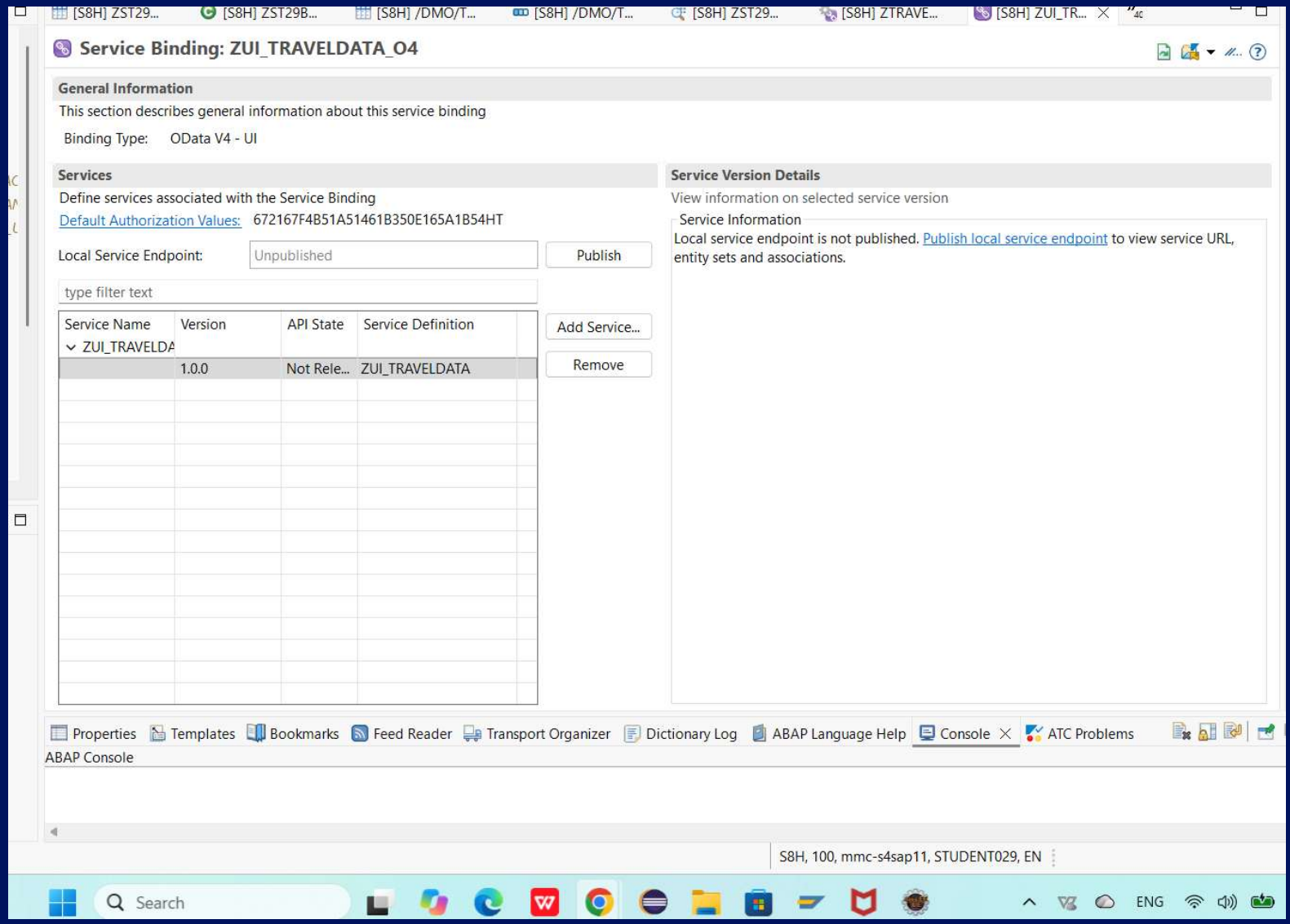

Browse...
ance Object
aintenance
TOL-MBC
Validation Status
Last update: Nov 11, 2026, 2:05:18 AM
Errors (1 Item)
Please set delivery class of database table ZST29_TRAVEL to C.

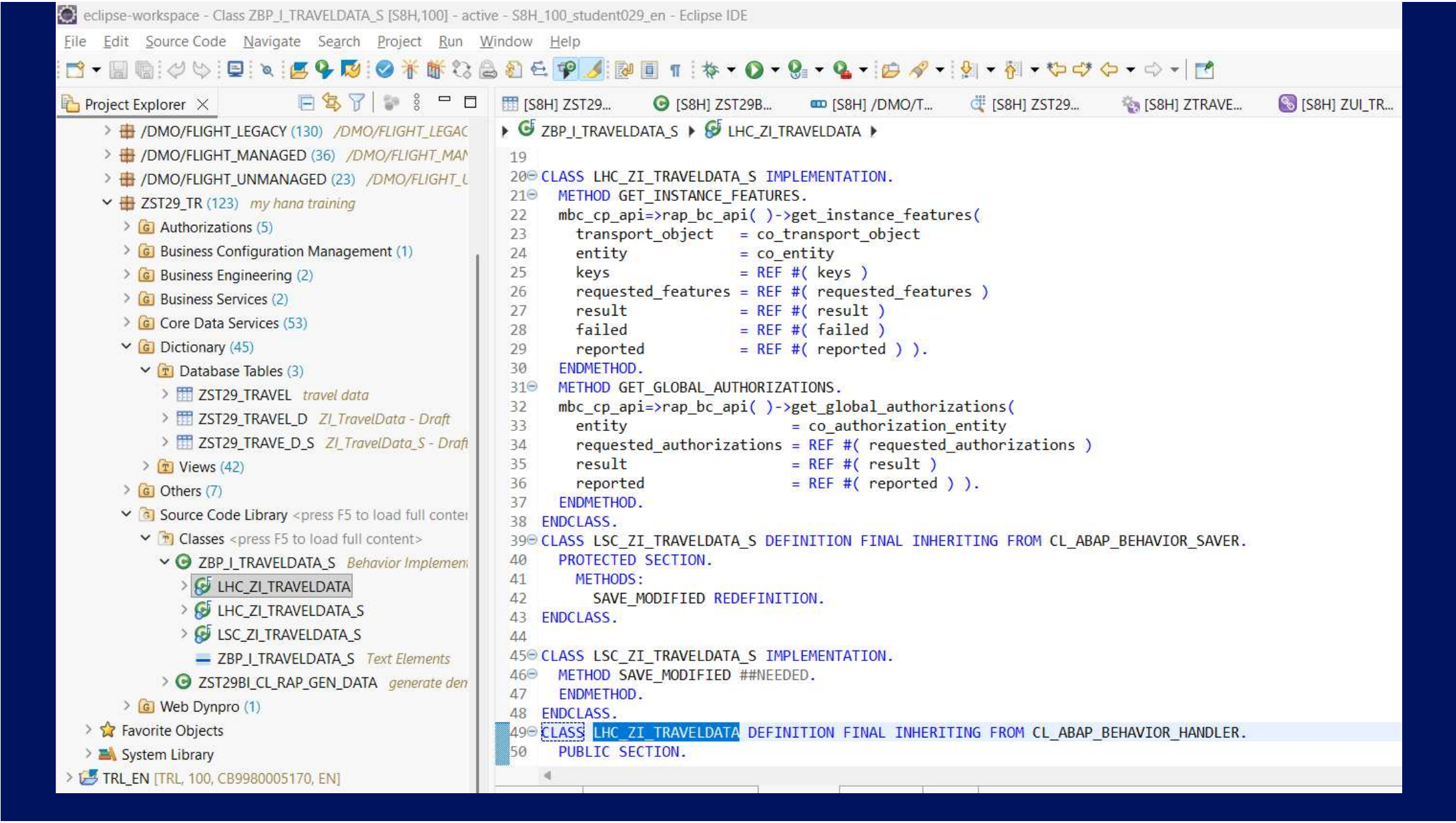

S8H(1)/100 Dictionary: Change Table
Table Edit Goto Utilities Extras Environment System Help
Technical Settings
Append Structure...
Transparent Table
ZST29_TRAVEL
Active
Short Description
travel data
Attributes
Delivery and Maintenance
Fields
Input Help/Check
Currency/Quantity Fields
Indexes
Delivery Class
C
Customizing, maintenance only by customer, no SAP imports
Data Browser/Table View Editing
Display/Maintenance Allowed with Restrictions

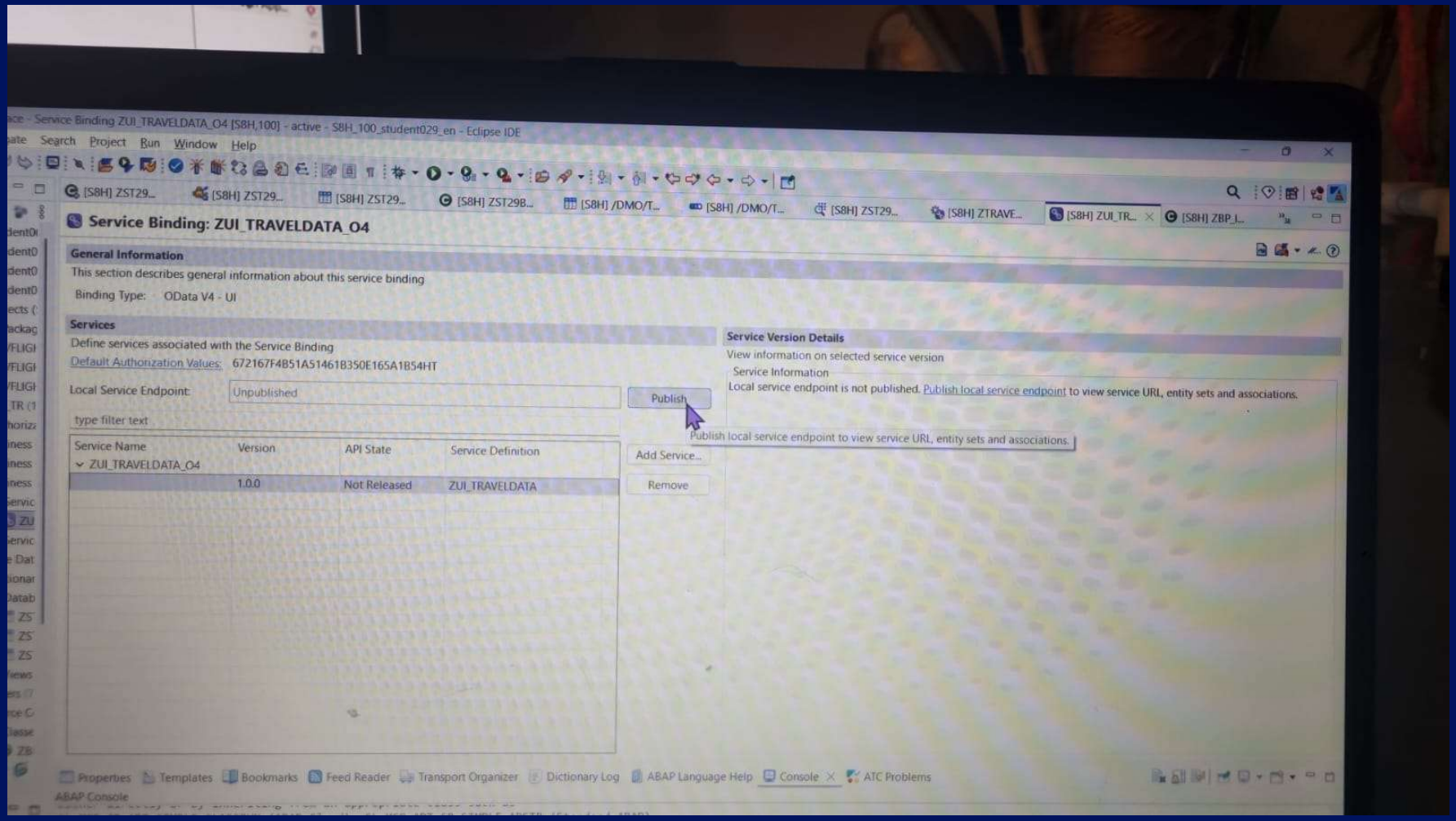

Configure Generator
Specify values for the generator, 'Business Configuration Maintenance Object'.
Business Configuration Application
type filter text
Scenario Options
BC Management
Business Service
Business Object
Data Model
Behavior Implementation
Scenario Options
Add Copy Action
Add Deprecate Actions
Add Data Consistency Check
Description:
Maintain travel data
Copy Parameter Entity:
ZD_CopyTravelDataP
Transport Selection:
No transport
Enable Transport Selection Strip
Create Projection Layer
< Back
Next >
Finish
Cancel

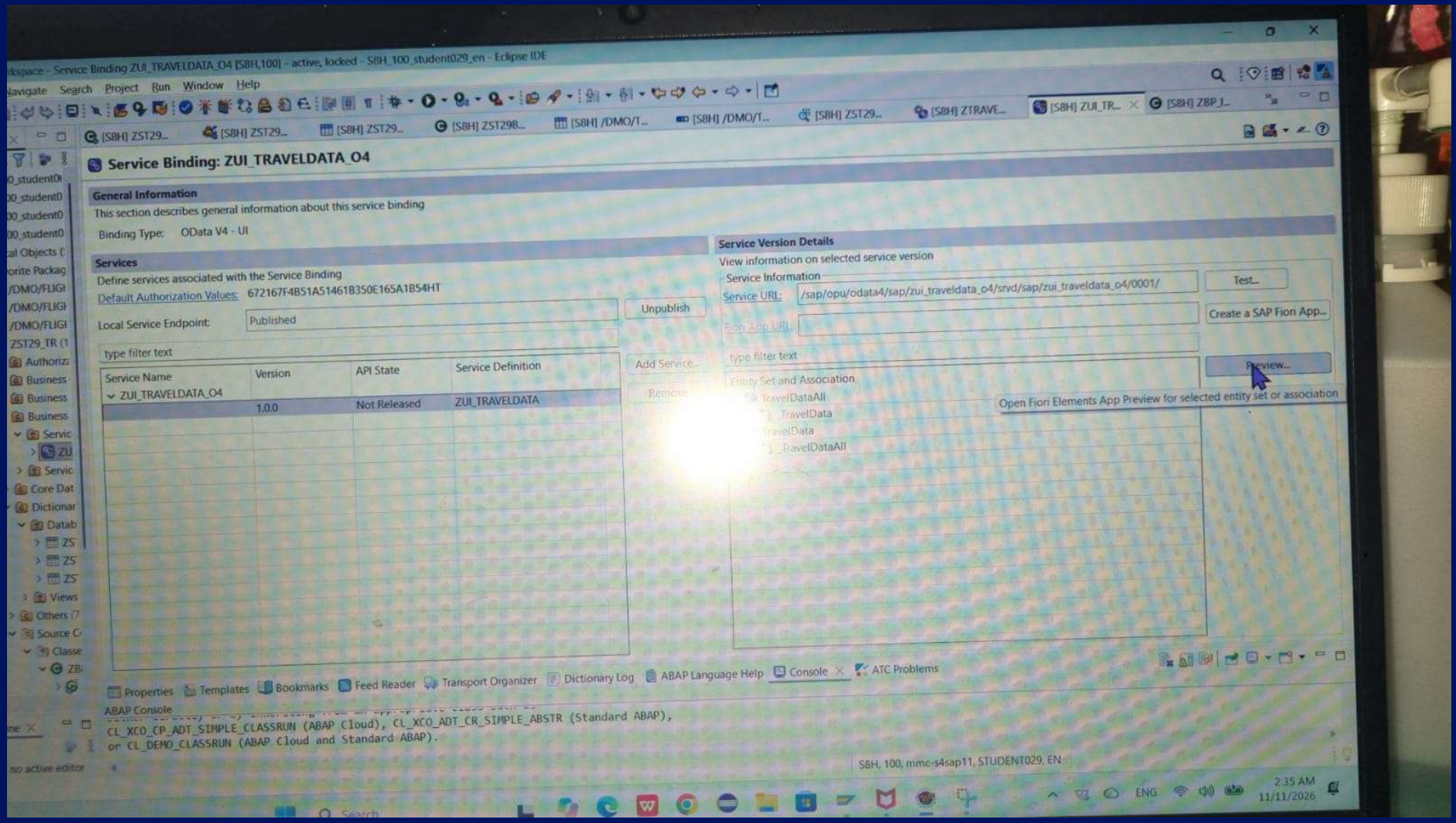

Generate ABAP Repository Objects
ABAP Artifacts Generation/Modification List
List of repository objects that are going to be generated.
type filter text
Repository Object Name
ABAP Object Type
Operation
Business Configuration Applica
BC Management
ZTRAVELDATA
Business Configuration Maintenanc...
CREATE
ZTRAVELDATA
Transport Object Definition
CREATE
Business Service
ZUI_TRAVELDATA_O4
Service Binding
CREATE
ZUI_TRAVELDATA
Service Definition
CREATE
Business Object
ZI_TRAVELDATA_S
Behavior Definition
CREATE
ZBP_I_TRAVELDATA_S
Class
CREATE
Data Model
ZI_TRAVELDATA_S
Data Definition
CREATE
ZI_TRAVELDATA
Data Definition
CREATE
ZI_TRAVELDATA
Metadata Extension
CREATE
ZI_TRAVELDATA
Access Control
CREATE
ZST29_TRAVEL_D
Database Table
CREATE
ZST29_TRAVE_D_S
Database Table
CREATE
Preview:
{
"formatVersion": "1 ",
"header": {
"description": "Maintain travel data",
"originalLanguage": "en"
},
"serviceConfiguration": {
"serviceBinding": "ZUI_TRAVELDATA_O4",
"serviceName": "ZUI_TRAVELDATA_O4",
"serviceVersion": "0001",
"rootEntitySet": "TravelDataAll",
< Back
Next >
Finish
Cancel
Data Preview
find pattern
CLIENT
Properties
ABAP Console

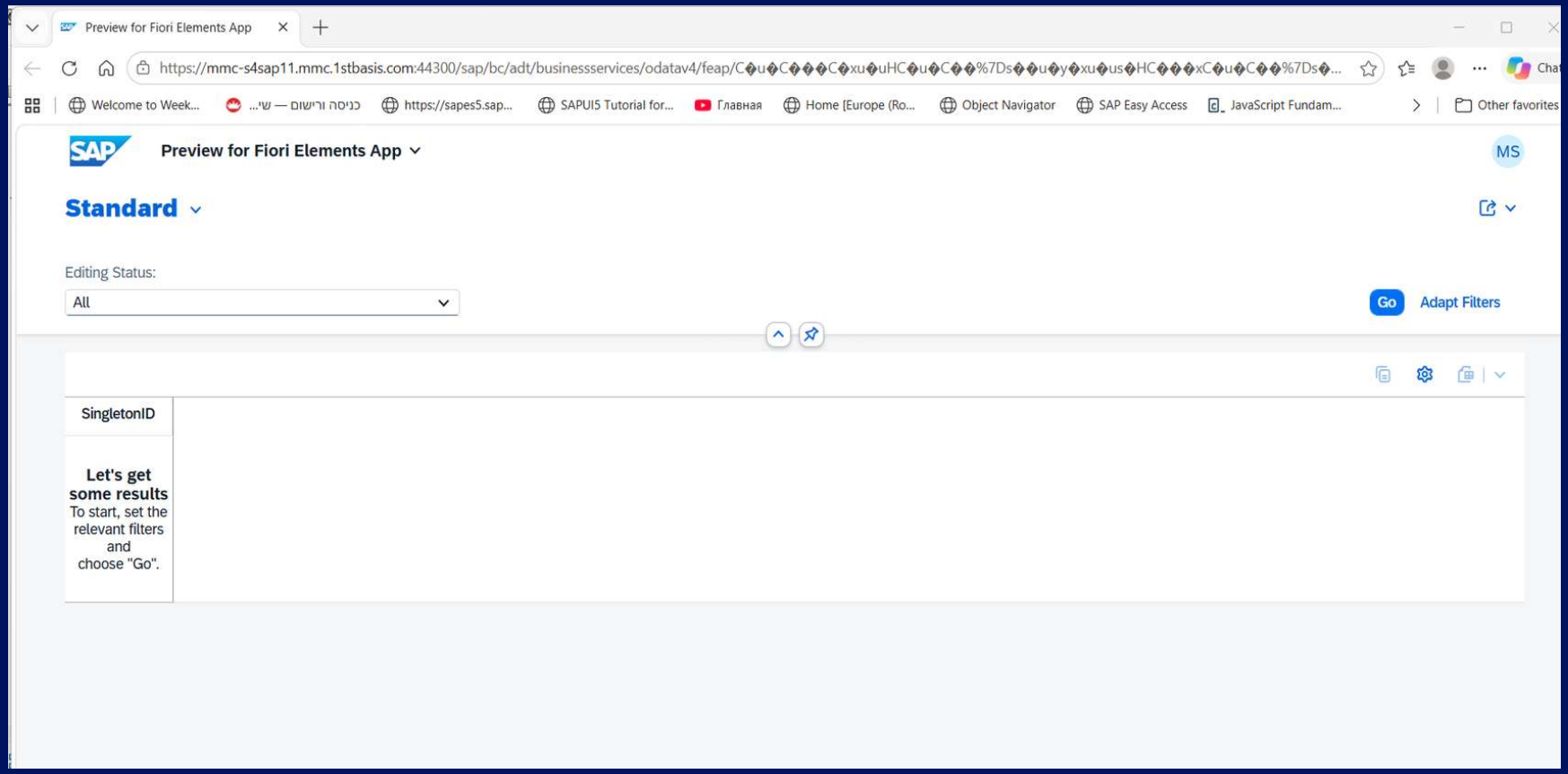

eclipse-workspace - Business Configuration Maintenance Object ZTRAVELDATA [S8H,100] - active - S8H_100_student029_en - Eclipse IDE
File Edit Navigate Search Project Run Window Help
Project Explorer
S23_100_student067_en [S23, 100, STUDENT067, EN]
S8H_100_student009_en [S8H, 100, STUDENT009, EN]
S8H_100_student009_en_1 [S8H, 100, STUDENT009, EN]
S8H_100_student029_en [S8H, 100, STUDENT029, EN]
Local Objects ($TMP)
Favorite Packages (312)
/DMO/FLIGHT_LEGACY (130) /DMO/FLIGHT_LEGAC
/DMO/FLIGHT_MANAGED (36) /DMO/FLIGHT_MAN
/DMO/FLIGHT_UNMANAGED (23) /DMO/FLIGHT_U
ZST29_TR (123) my hana training
Authorizations (5)
Business Configuration Management (1)
Business Engineering (2)
Business Services (2)
Core Data Services (53)
Dictionary (45)
Database Tables (3)
ZST29_TRAVEL travel data
ZST29_TRAVEL_D ZI_TravelData - Draft
Outline
There is no active editor that provides an outline.
[S8H] ZST29...
[S8H] ZST29...
[S8H] ZST29B...
[S8H] /DMO/T...
[S8H] /DMO/T...
[S8H] ZST29...
[S8H] ZTRAVE...
Business Configuration Maintenance Object: ZTRAVELDATA
Service Configuration
Service Binding: * ZUI_TRAVELDATA_O4 Browse...
Service Name: * ZUI_TRAVELDATA_O4 Browse...
Service Version: * 0001 Browse...
Root Entity Set: * TravelDataAll Browse...
Name: * travel data
Configuration Group: Browse...
Skip Root Entity List Report
List Report Configuration
Initial Load: Enabled
Variant Management: Page
Object Page Configurations
Entity Set | Section Layout | Variant Management | Editable Header Cont... | Show Transport Selec...
TravelDataAll | Page | Control | No | No
Add...
Edit...
Remove
Properties Templates Bookmarks Feed Reader Transport Organizer Dictionary Log ABAP Language Help Console ATC Problems
ABAP Console
'Business Configuration Maintenance Object' generated the artifacts. Refresh (F5) the package to view the objects.
S8H, 100, mmc-s4sap11, STUDENT029, EN

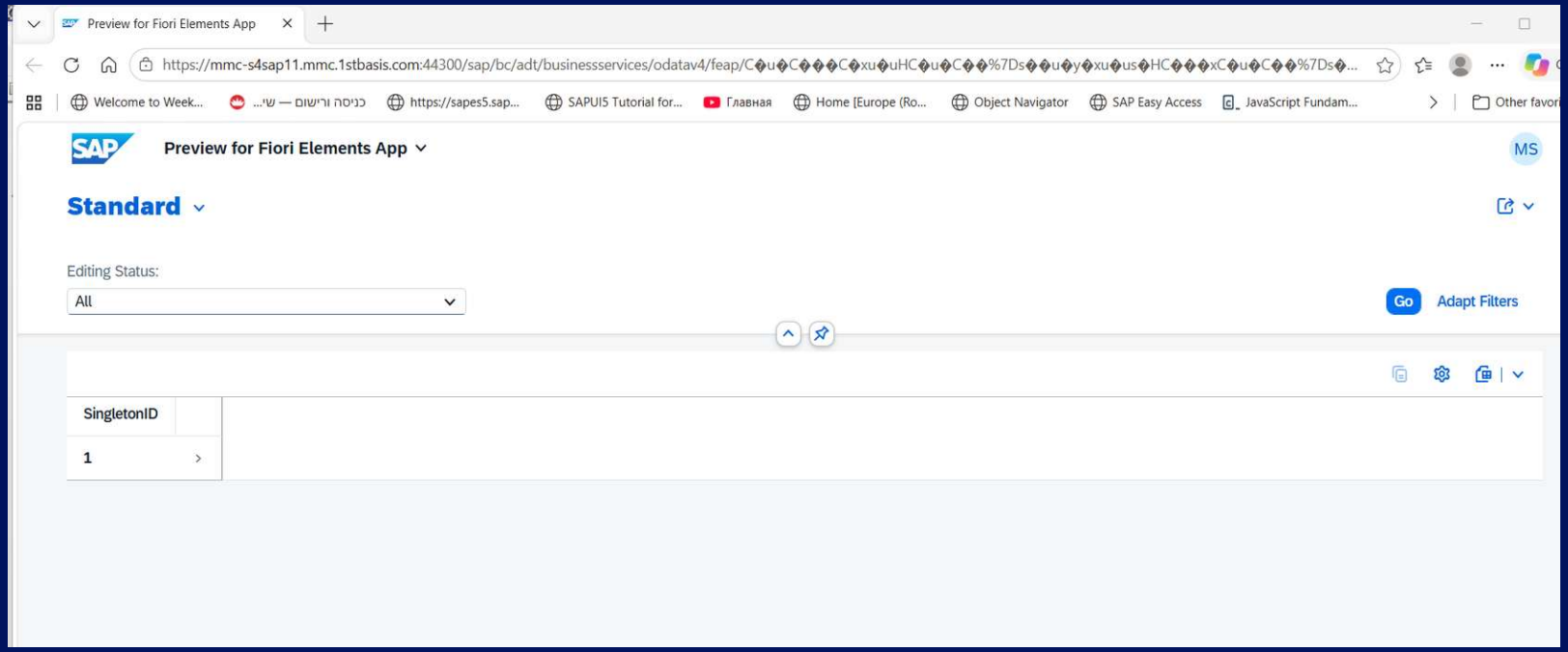

[S8H] ZST29...
[S8H] ZST29B...
[S8H] /DMO/T...
[S8H] /DMO/T...
[S8H] ZST29...
[S8H] ZTRAVE...
[S8H] ZUI_TR...
Service Binding: ZUI_TRAVELDATA_O4
General Information
This section describes general information about this service binding
Binding Type: OData V4 - UI
Services
Define services associated with the Service Binding
Default Authorization Values: 672167F4B51A51461B350E165A1B54HT
Local Service Endpoint: Unpublished
Publish
type filter text
Service Name
Version
API State
Service Definition
ZUI_TRAVELDA
1.0.0
Not Rele...
ZUI_TRAVELDATA
Add Service...
Remove
Service Version Details
View information on selected service version
Service Information
Local service endpoint is not published. Publish local service endpoint to view service URL, entity sets and associations.
Properties
Templates
Bookmarks
Feed Reader
Transport Organizer
Dictionary Log
ABAP Language Help
Console
ATC Problems
ABAP Console
S8H, 100, mmc-s4sap11, STUDENT029, EN
Search
ENG

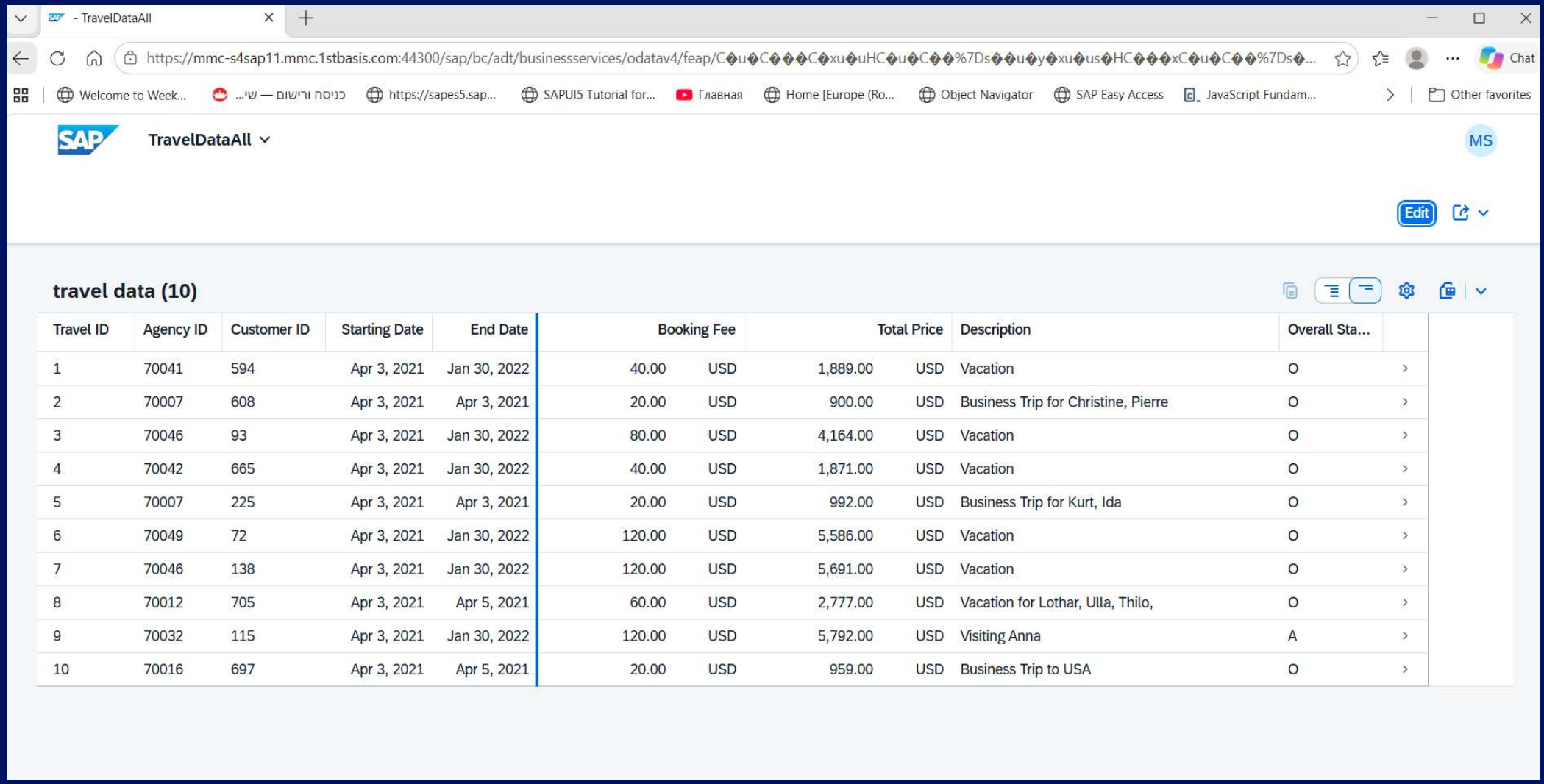

eclipse-workspace - Class ZBP_I_TRAVELDATA_S [S8H,100] - active - S8H_100_student029_en - Eclipse IDE
File Edit Source Code Navigate Search Project Run Window Help
Project Explorer
/DMO/FLIGHT_LEGACY (130) /DMO/FLIGHT_LEGAC
/DMO/FLIGHT_MANAGED (36) /DMO/FLIGHT_MAN
/DMO/FLIGHT_UNMANAGED (23) /DMO/FLIGHT_U
ZST29_TR (123) my hana training
Authorizations (5)
Business Configuration Management (1)
Business Engineering (2)
Business Services (2)
Core Data Services (53)
Dictionary (45)
Database Tables (3)
ZST29_TRAVEL travel data
ZST29_TRAVEL_D ZI_TravelData - Draft
ZST29_TRAVE_D_S ZI_TravelData_S - Draft
Views (42)
Others (7)
Source Code Library <press F5 to load full conte
Classes <press F5 to load full content>
ZBP_I_TRAVELDATA_S Behavior Implemen
LHC_ZI_TRAVELDATA
LHC_ZI_TRAVELDATA_S
LSC_ZI_TRAVELDATA_S
ZBP_I_TRAVELDATA_S Text Elements
ZST29BI_CL_RAP_GEN_DATA generate den
Web Dynpro (1)
Favorite Objects
System Library
TRL_EN [TRL, 100, CB9980005170, EN]
[S8H] ZST29...
[S8H] ZST29B...
[S8H] /DMO/T...
[S8H] ZST29...
[S8H] ZTRAVE...
[S8H] ZUI_TR...
ZBP_I_TRAVELDATA_S
LHC_ZI_TRAVELDATA
19
20 CLASS LHC_ZI_TRAVELDATA_S IMPLEMENTATION.
21 METHOD GET_INSTANCE_FEATURES.
22 mbc_cp_api=>rap_bc_api( )->get_instance_features(
23 transport_object = co_transport_object
24 entity = co_entity
25 keys = REF #(keys)
26 requested_features = REF #(requested_features)
27 result = REF #(result)
28 failed = REF #(failed)
29 reported = REF #(reported)).
30 ENDMETHOD.
31 METHOD GET_GLOBAL_AUTHORIZATIONS.
32 mbc_cp_api=>rap_bc_api( )->get_global_authorizations(
33 entity = co_authorization_entity
34 requested_authorizations = REF #(requested_authorizations)
35 result = REF #(result)
36 reported = REF #(reported)).
37 ENDMETHOD.
38 ENDCLASS.
39 CLASS LSC_ZI_TRAVELDATA_S DEFINITION FINAL INHERITING FROM CL_ABAP_BEHAVIOR_SAVER.
40 PROTECTED SECTION.
41 METHODS:
42 SAVE_MODIFIED REDEFINITION.
43 ENDCLASS.
44
45 CLASS LSC_ZI_TRAVELDATA_S IMPLEMENTATION.
46 METHOD SAVE_MODIFIED ##NEEDED.
47 ENDMETHOD.
48 ENDCLASS.
49 CLASS LHC_ZI_TRAVELDATA DEFINITION FINAL INHERITING FROM CL_ABAP_BEHAVIOR_HANDLER.
50 PUBLIC SECTION.


Service Binding ZUI_TRAVELDATA_O4 [S8H,100] - active - S8H_100_student029_en - Eclipse IDE
Search Project Run Window Help
Service Binding: ZUI_TRAVELDATA_O4
General Information
This section describes general information about this service binding
Binding Type: OData V4 - UI
Services
Define services associated with the Service Binding
Default Authorization Values: 672167F4B51A51461B350E165A1B54HT
Local Service Endpoint: Unpublished
Publish
type filter text
Service Name
Version
API State
Service Definition
ZUI_TRAVELDATA_O4
1.0.0
Not Released
ZUI_TRAVELDATA
Add Service...
Remove
Service Version Details
View information on selected service version
Service Information
Local service endpoint is not published. Publish local service endpoint to view service URL, entity sets and associations.
Publish local service endpoint to view service URL, entity sets and associations.
Properties
Templates
Bookmarks
Feed Reader
Transport Organizer
Dictionary Log
ABAP Language Help
Console
ATC Problems
ABAP Console


Service Binding: ZUI_TRAVELDATA_O4
General Information
This section describes general information about this service binding
Binding Type: OData V4 - UI
Services
Define services associated with the Service Binding
Default Authorization Values: 672167F4B51A51461B350E165A1B54HT
Local Service Endpoint: Published
Unpublish
type filter text
Service Name
Version
API State
Service Definition
ZUI_TRAVELDATA_O4
1.0.0
Not Released
ZUI_TRAVELDATA
Add Service...
Remove
Service Version Details
View information on selected service version
Service Information
Service URL: /sap/opu/odata4/sap/zui_traveldata_o4/srvd/sap/zui_traveldata_o4/0001/
Test...
Create a SAP Fiori App...
Preview...
Entity Set and Association
TravelDataAll
_TravelData
TravelData
_TravelDataAll
Open Fiori Elements App Preview for selected entity set or association
Properties
Templates
Bookmarks
Feed Reader
Transport Organizer
Dictionary Log
ABAP Language Help
Console
ATC Problems
ABAP Console
CL_XCO_CP_ADT_SIMPLE_CLASSRUN (ABAP Cloud), CL_XCO_ADT_CR_SIMPLE_ABSTR (Standard ABAP),
or CL_DEMO_CLASSRUN (ABAP Cloud and Standard ABAP).
S8H, 100, mmc-s4sap11, STUDENT029, EN
2:35 AM
11/11/2026


Preview for Fiori Elements App
https://mmc-s4sap11.mmc.1stbasis.com:44300/sap/bc/adt/businessservices/odatav4/feap/C%u0C%u0C%u0C%xu%uHC%u0C%%7Ds%%u%y%xu%us%HC%%%xC%u0C%%7Ds%...
Welcome to Week...
כניסה ורישום — שי...
https://sapes5.sap...
SAPUI5 Tutorial for...
Главная
Home [Europe (Ro...
Object Navigator
SAP Easy Access
JavaScript Fundam...
Other favorites
Preview for Fiori Elements App
MS
Standard
Editing Status:
All
Go
Adapt Filters
SingletonID
Let's get some results
To start, set the relevant filters and choose "Go".

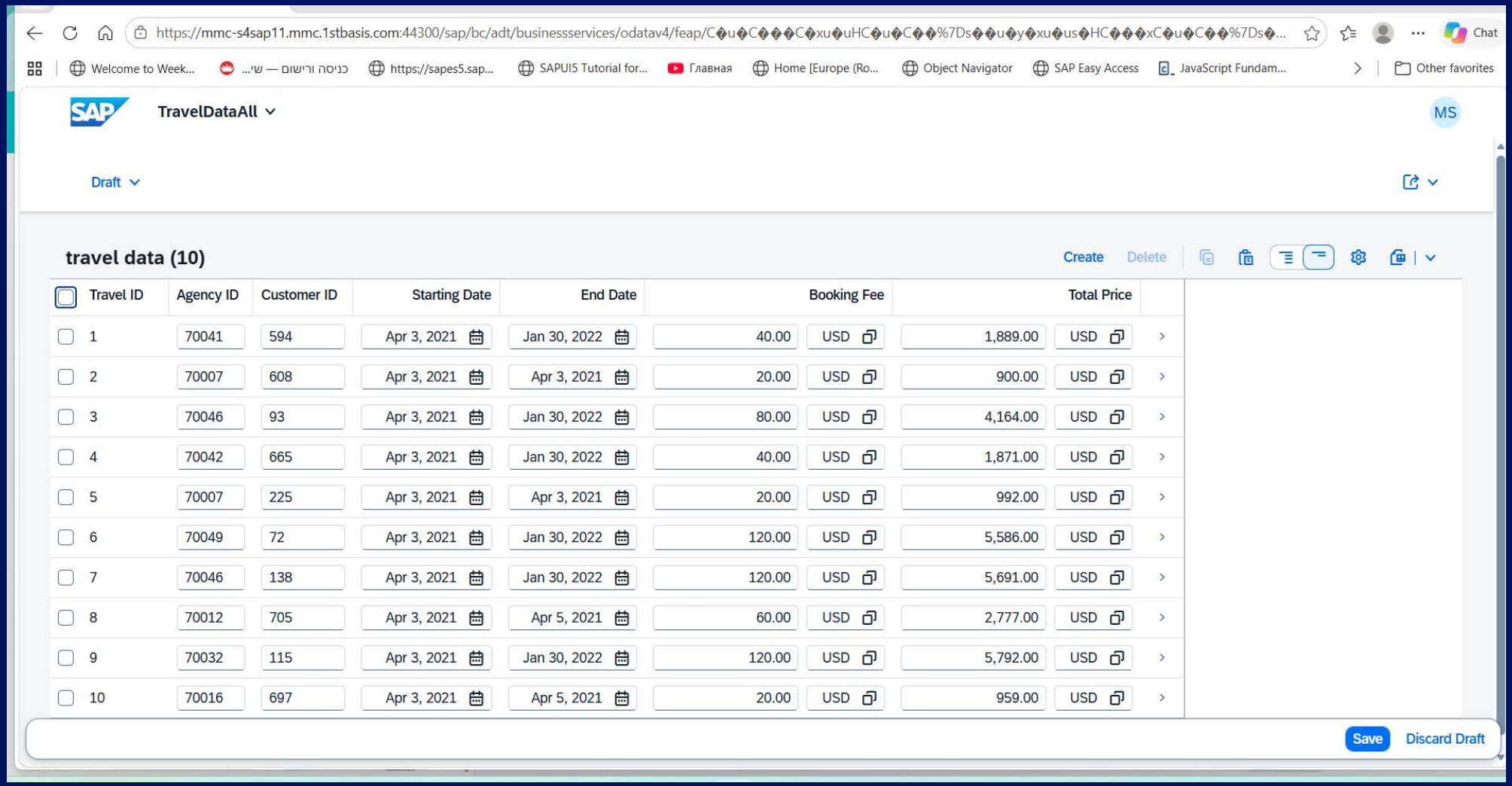

Preview for Fiori Elements App
https://mmc-s4sap11.mmc.1stbasis.com:44300/sap/bc/adt/businessservices/odatav4/feap/C%u0C%u0C%u0C%xu%uHC%u0C%%7Ds%%u%y%xu%us%HC%%%xC%u0C%%7Ds%...
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Preview for Fiori Elements App
MS
Standard
Editing Status:
All
Go
Adapt Filters
SingletonID
1



travel data (10)

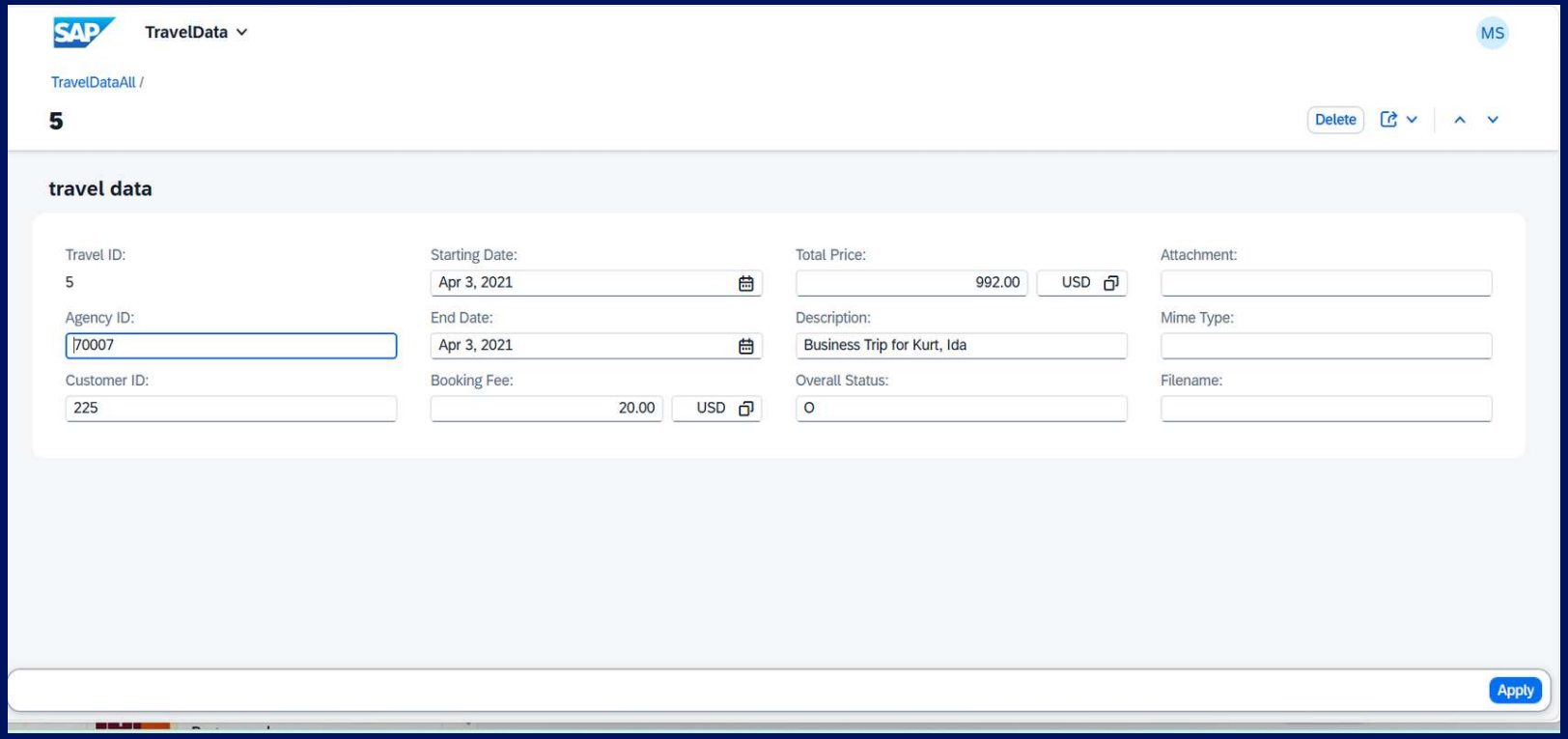
Travel ID	Agency ID	Customer ID	Starting Date	End Date	Booking Fee		Total Price		Description	Overall Sta...
1	70041	594	Apr 3, 2021	Jan 30, 2022	40.00	USD	1,889.00	USD	Vacation	O
2	70007	608	Apr 3, 2021	Apr 3, 2021	20.00	USD	900.00	USD	Business Trip for Christine, Pierre	O
3	70046	93	Apr 3, 2021	Jan 30, 2022	80.00	USD	4,164.00	USD	Vacation	O
4	70042	665	Apr 3, 2021	Jan 30, 2022	40.00	USD	1,871.00	USD	Vacation	O
5	70007	225	Apr 3, 2021	Apr 3, 2021	20.00	USD	992.00	USD	Business Trip for Kurt, Ida	O
6	70049	72	Apr 3, 2021	Jan 30, 2022	120.00	USD	5,586.00	USD	Vacation	O
7	70046	138	Apr 3, 2021	Jan 30, 2022	120.00	USD	5,691.00	USD	Vacation	O
8	70012	705	Apr 3, 2021	Apr 5, 2021	60.00	USD	2,777.00	USD	Vacation for Lothar, Ulla, Thilo,	O
9	70032	115	Apr 3, 2021	Jan 30, 2022	120.00	USD	5,792.00	USD	Visiting Anna	A
10	70016	697	Apr 3, 2021	Apr 5, 2021	20.00	USD	959.00	USD	Business Trip to USA	O

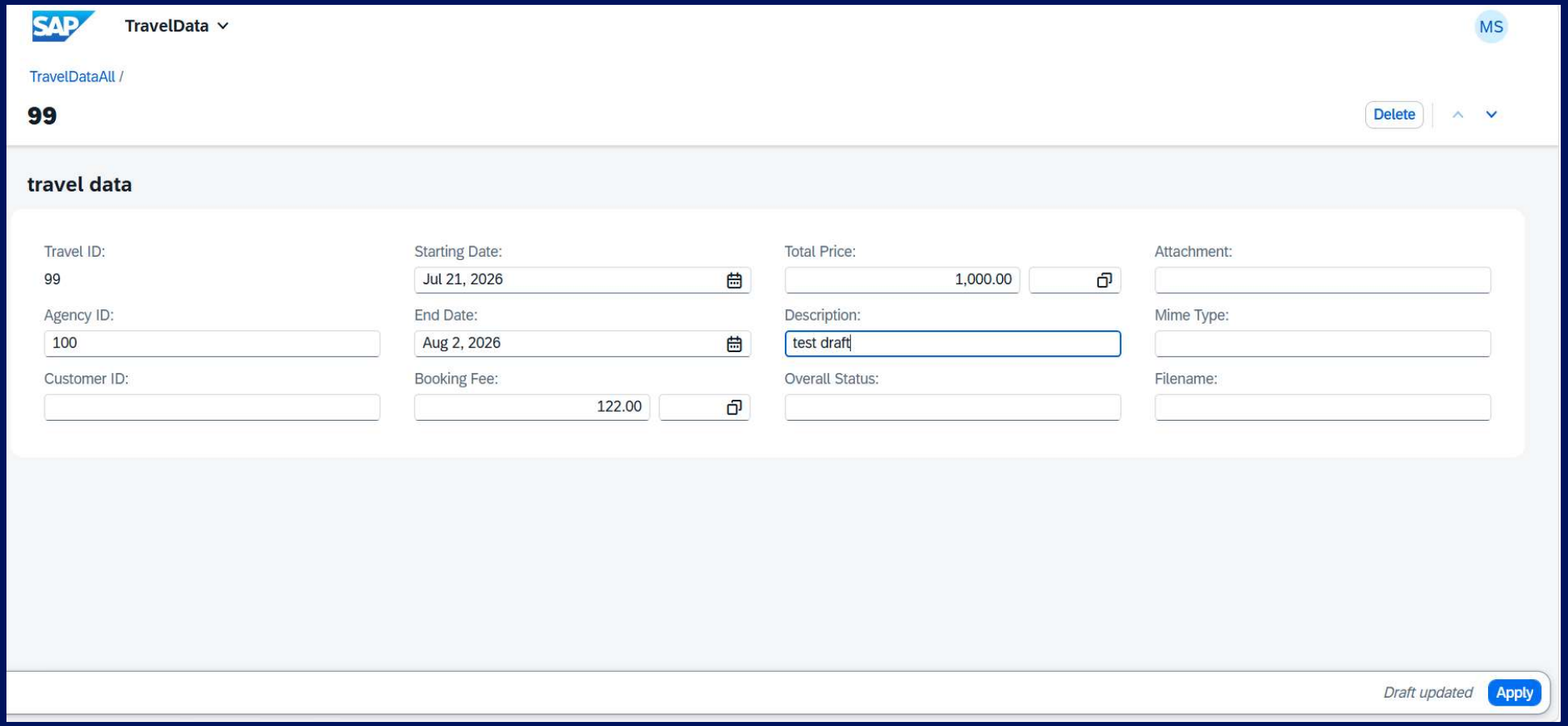
PUSH ON EDIT

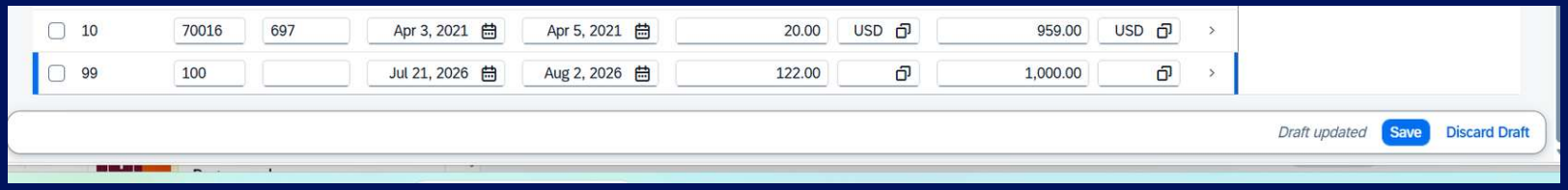


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10	70016	697	Apr 3, 2021	Apr 5, 2021	20.00	USD	959.00	USD

WE MAY CHOOSE ANY ROW AND EDIT DATA OR ADD NEW ROWS




TravelData
MS
TravelDataAll /
99
Delete
travel data
Travel ID:
99
Starting Date:
Jul 21, 2026
Total Price:
1,000.00
Attachment:
Agency ID:
100
End Date:
Aug 2, 2026
Description:
test draft
Mime Type:
Customer ID:
Booking Fee:
122.00
Overall Status:
Filename:
Draft updated
Apply


10 70016 697 Apr 3, 2021 Apr 5, 2021 20.00 USD 959.00 USD
99 100 Jul 21, 2026 Aug 2, 2026 122.00 1,000.00
Draft updated
Save
Discard Draft

TO BE CONTINUED